

Shiye (Sally) Cao

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RESEARCH OVERVIEW

I develop conversational capabilities that enable robots to communicate naturally, effectively, and robustly as they work alongside people in their daily lives.

Keywords: human-robot interaction (HRI), embodied AI, privacy and security, education.

EDUCATION

Johns Hopkins University, Baltimore, Maryland, USA

PhD Candidate in Computer Science advised by [Chien-Ming Huang](#) and [Anqi Liu](#) 2022–Present

M.S.E in Computer Science 2021–2022

B.S. in Computer Science and Applied Mathematics and Statistics 2018–2021

PUBLICATIONS

*Indicates authors contribute equally to the work

All publications can be found on my [Google Scholar](#) profile. Top-tier conferences (HRI, RSS, CHI, CSCW) have acceptance rates around 25% and are considered premier venues for Computer Science research.

Peer-Reviewed Conference Papers

C.7 **ERR@ HRI 3.0 Challenge: Multimodal Detection of Errors and Anticipation in Human-Robot Interactions**

Maria Teresa Parreira, Micol Spitale, Maia Stiber, **Shiye Cao**, Amama Mahmood, Chien-Ming Huang, Hatice Gunes, Wendy Ju

Proceedings of the 28th International Conference on Multimodal Interaction (ICMI'26)

<https://arxiv.org/abs/2607.11570>

C.6 **ELLA: Generative AI-Powered Social Robots for Early Language Development at Home**

Victor Nikhil Antony*, **Shiye Cao***, Shuning Wang, and Chien-Ming Huang

Proceedings of the 25th Annual ACM Interaction Design and Children Conference (IDC'26)

<https://dl.acm.org/doi/10.1145/3773077.3806113>

C.5 **Reframing Conversational Design in HRI: Deliberate Design with AI Scaffolds**

Shiye Cao, Jiwon Moon, Yifan Xu, Anqi Liu, and Chien-Ming Huang

Proceedings of 2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI'26) | Acceptance rate: 23.2%

<https://dl.acm.org/doi/10.1145/3757279.3785640>

C.4 **ERR@ HRI 2.0 Challenge: Multimodal Detection of Errors and Failures in Human-Robot Conversations**

Shiye Cao, Maia Stiber, Amama Mahmood, Maria Teresa Parreira, Wendy Ju, Micol Spitale, Hatice Gunes, and Chien-Ming Huang

Proceedings of 33rd ACM International Conference on Multimedia (ACM-MM'25)

<https://dl.acm.org/doi/abs/10.1145/3746027.3762073>

C.3 **Interruption Handling for Conversational Robots**

Shiye Cao*, Jiwon Moon*, Amama Mahmood, Victor Nikhil Antony, Ziang Xiao, Anqi Liu, and

Chien-Ming Huang

Proceedings of the 2025 Robotics: Science and Systems Conference (RSS'25)

<https://www.roboticsproceedings.org/rss21/p089.pdf> | Acceptance rate: 27.4%

C.2 **Voice Assistants for Health Self-Management: Designing for and with Older Adults**

Amama Mahmood, **Shiye Cao**, Maia Stiber, Victor Nikhil Antony, and Chien-Ming Huang

Proceedings of 2025 ACM Conference on Human Factors in Computing Systems (CHI'25)

<https://dl.acm.org/doi/10.1145/3706598.3713839> | Acceptance rate: 25.1%

C.1 **“What If It Is Wrong”: Effects of Power Dynamics and Trust Repair Strategy on Trust and Compliance in HRI**

Ulas Berk Karli*, **Shiye Cao***, and Chien-Ming Huang

Proceedings of 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI'23)

<https://dl.acm.org/doi/10.1145/3568162.3576964> | Acceptance rate: 25.3%

Peer-Reviewed Journal Articles

J.4 **Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making**

Shiye Cao, Anqi Liu, and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'24)

Volume 8, Issue CSCW1, Article 41, Pages 1-32

<https://dl.acm.org/doi/10.1145/3637318>

J.3 **How Time Pressure from Different Phases of Decision-Making Influences Human-AI Collaboration**

Shiye Cao*, Catalina Gomez*, and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'23)

Volume 7, Issue CSCW2, Article 277, Pages 1-26

<https://dl.acm.org/doi/10.1145/3610068>

J.2 **Crowdsourcing Thumbnail Captions: Data Collection and Validation**

Carlos Aguirre*, **Shiye Cao***, Amama Mahmood, and Chien-Ming Huang

ACM Transactions on Interactive Intelligent Systems (TiiS, 2023)

Volume 13, Issue 3, Article 14, Pages 1-28

<https://dl.acm.org/doi/10.1145/3589346>

J.1 **Understanding User Reliance on AI in Assisted Decision-Making**

Shiye Cao and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'22)

Volume 6, Issue CSCW2, Article 471, Pages 1-23

<https://dl.acm.org/doi/10.1145/3555572>

Refereed Symposium & Workshop Papers

W.4 **Rapid Conversational Authoring of Interactive Human-Robot Activities**

Shiye Cao*, Aureliane Tchamdja*, Chien-Ming Huang

To appear at the AAAI Fall Symposium on Unifying Representations for Robot Application Development (UR-RAD'26)

- W.3 **Long-Term In-Home Robotic Support for Children with ASD**
Shiye Cao
 2026 HRI Pioneer Workshop | Acceptance rate: 23%
<https://dl.acm.org/doi/10.1145/3776734.3794629>
- W.2 **Practical Trade-offs in Generative Robot Expression Design**
 Victor Nikhil Antony, **Shiye Cao**, Chien-Ming Huang
 Expressive robot faces with Vizij: a workshop / tutorial hybrid at HRI 2026
<https://drive.google.com/file/d/1oZPRrEJ829Ju4gAeAPaSNcZOqEjEAp9b/view>
- W.1 **Eyes Are the Windows to AI Reliance: Towards Real-Time Human-AI Reliance Assessment**
Shiye Cao, Shichang Ke*, Alexandra Mo*, Anqi Liu, and Chien-Ming Huang
 2023 CHI Workshop on Trust and Reliance in AI-Assisted Tasks (TRAIT'23)
https://chi-trait.github.io/papers/2023/CHI_TRAIT_2023_Paper_35.pdf

Under-Review Papers

- U.3 **Privacy and Security Communication for Robots: Current Practice, Open Challenges, and Recommendations**
Shiye Cao, Laura Fleig, Yaxing Yao, Chien-Ming Huang
- U.2 **Towards Child-Centered Error Management In-the-Wild: Children's Reactions and Recovery Following Robot Errors**
Shiye Cao*, Shuning Wang*, Victor Nikhil Antony, Leyun Kim, Chien-Ming Huang
- U.1 **Sync or Sink: Conversational Pace Matching for Human-Robot Collaboration**
 Yifan Yin*, Binze Li*, **Shiye Cao***, Katherine Guerrerio, Seokhyun Baek, Tianmin Shu, and Chien-Ming Huang

AWARDS

Best Poster Award, Third Annual Celebrating Women in Data Science and AI Symposium	2026
HRI Pioneer	2026
CRA Outstanding Undergraduate Researcher Finalist	2022

TEACHING EXPERIENCES

Teaching Assistant, Introduction to Human-Robot Interaction	Spring 2026
Teaching Assistant, Introduction to Human-Computer Interaction	Fall 2022
Course Assistant, Machine Learning	Fall 2020, Spring 2021, Fall 2021

PROFESSIONAL SERVICE

Organizing Committee: ICMI'26 ERR@HRI 3.0 Challenge, ACM-MM'25 ERR@HRI 2.0 Challenge
Program Committee: 2027 HRI Pioneers Workshop, NeurIPS '22 Gaze Meets ML Workshop
Reviewer: HRI'26, CHI'26 (Outstanding Review), CSCW'25, CHI'25, HRI'25, HRI'24 (Outstanding Review), CSCW'24, CSCW'23, THRI
Student Volunteer: HRI'23, CSCW'23, CHI'23, CSCW'22
Community Outreach:

- Organized outreach event at the Applied ABC Autism Therapy Center (4/2026)
- Organized outreach event at the Port Discovery Children's Museum (6/2026)

- Organized outreach event at the Henderson Hopkins School (8/2026)

RESEARCH MENTORING

I have mentored nine students on projects leading to publications. Mentees have continued to PhD programs at Yale and the University of Chicago, and one received a CRA Outstanding Undergraduate Researcher Honorable Mention.

PhD Students

Yifan Yin (JHU CS PhD)	2026
Publications: U.1	
Binze Li (JHU CS PhD)	2026
Publications: U.1	
Laura Fleig (JHU CS PhD)	2026
Publication: U.3	

Master's Students

Ulas Berk Karli (JHU Robotics MSE)	2021–2023
Next: CS PhD at Yale University	
Publications: C.1	
Yifan Xu (JHU CS MSE)	2025–2026
Publications: C.5	
Shuning Wang (JHU Systems Engineering MSE)	2025–2026
Publications: C.6, U.2	

Undergraduate Students

Jiwon Moon (JHU CS BS)	2023–2025
Next: CS PhD at the University of Chicago	
 2025 CRA Outstanding Undergraduate Researcher Honorable Mention	
Publications: C.3, C.5	
Leyun Kim (JHU CS & Cognitive Science BS)	2026
Publications: U.2	
Aureliane Tchamdja (UMBC CS BS, NSF REU)	2026
Publications: W.4	

Selected Press

[Celebrating women in data science and AI](#) (JHU CS News)

[Aging gracefully with AI](#) (JHU CS News)

[Talking robots learn to manage human interruptions](#) (JHU HUB)

[CS student named finalist for 2022 CRA Outstanding Undergraduate Researcher Award](#) (JHU CS News)